SeerRAG: Seer-like Routing for Uncertainty-guided Retrieval-Augmented Generation

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***Abstract*—Retrieval-Augmented Generation (RAG) has emerged as an effective paradigm for mitigating factual hallucinations in Large Language Models (LLMs).However, existing RAG systems encounter two fundamental challenges when processing heterogeneous multimodal long documents: (1) indiscriminate retrieval triggering, which invokes retrieval for all queries regardless of necessity, leading to excessive computational overhead and noisy evidence; and (2) rigid modality mapping, where static retrieval strategies fail to adapt to multi-granularity queries in mixed-modality scenarios.We propose SeerRAG, a dynamic routing framework featuring a "Perceive-Shunt-Coordinate" architecture. It integrates an Uncertainty Navigator (UN) for adaptive retrieval gating, a Modality Discrimination (MD) engine for semantic-aware path routing (textual/visual/hybrid), and a Cross-modal Synergistic Generation (CMS) module for deep evidence alignment. Experimental results on the LongDocURL benchmark show that SeerRAG boosts the accuracy of Qwen2-VL-7B and LLaVA-OneVision-Chat-7B to 67.3% and 55.1%, respectively, achieving a relative gain of ~120%. Furthermore, in a hybrid evaluation of complex and factual queries, SeerRAG effectively filters 97% of redundant triggers for the latter, confirming the precision of its uncertainty-based gating mechanism. Our work achieves a Pareto-optimal balance between accuracy and efficiency, providing a robust solution for multimodal long-document understanding.**

Keywords*—*Retrieval-augmented generation, multimodal RAG, dynamic retrieval routing, uncertainty estimation

I. INTRODUCTION

Recently, Large Language Models (LLMs) have demonstrated exceptional capabilities in context understanding and knowledge synthesis[1],[2]. However, these models rely heavily on parametric memory formed during pre-training, leading to inherent limitations such as factual hallucinations where LLMs generate confident but factually incorrect or unsupported content due to limited or outdated parametric knowledge[3]-and knowledge obsolescence when handling specialized or real-time information[4].To address these limitations, Retrieval-Augmented Generation (RAG) hasemerged as an effective paradigm by integrating non-parametric external knowledge bases, significantly enhancing accuracy and reliability in knowledge-intensive tasks [5], [6].

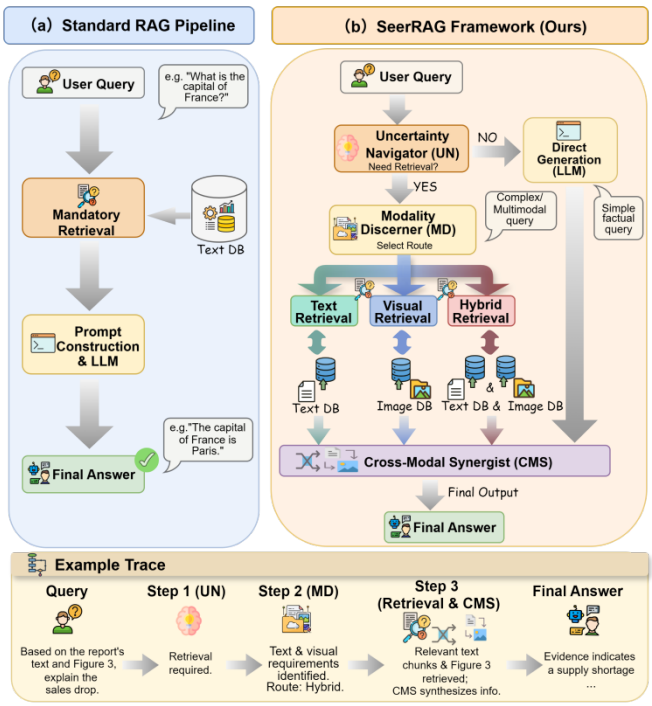


Fig. 1. Comparison of system architectures:a) Standard RAG: Executes mandatory retrieval for all queries regardless of complexity. b) SeerRAG: A hierarchical routing framework for intelligent scheduling. The Uncertainty Navigator (UN) first determines retrieval necessity; subsequently, the Modality Discerner (MD) adapts the retrieval path (Textual/Visual/Hybrid). The example trace illustrates the end-to-end workflow from uncertainty assessment to evidence synthesis for a sales analysis task.

With the rise of multimodal learning, traditional text-only RAG has been extended to the multimodal domain[7]. This transition is primarily motivated by the inherent requirements of long-document understanding tasks, including scientific literature analysis, financial report interpretation, and medical record review, which naturally involve complex layouts and heterogeneous information sources[8],[9]. Recent Large Vision-Language Models (LVLMs), such as Qwen2-VL[10], LLaVA-OneVision[11], and Gemini-1.5[12], are capable of directly processing heterogeneous text-image inputs.

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Nevertheless, when applied to long documents with intricate layouts and strong cross-modal dependencies, these models remain constrained by limited context window capacities and insufficient multimodal feature alignment, thereby hindering their effectiveness in such scenarios [10],[13].

Despite substantial progress in multimodal Retrieval-Augmented Generation (RAG), existing systems continue to face two critical bottlenecks that impede practical deployment [14],[15]. First, indiscriminate retrieval triggering: most systems adopt a full-retrieval paradigm that uniformly invokes external retrieval for all queries, incurring unnecessary computational overhead and potentially degrading generation quality by introducing irrelevant evidence[6],[14],[16]. Second, the lack of modality-aware routing: prior work predominantly focuses on single-modality tasks and fails to dynamically select appropriate modalities for mixed multimodal queries. Consequently, systems either default to textual retrieval, thereby overlooking informative visual cues in figures and tables, or forcibly trigger visual retrieval for text-dominant queries, leading to suboptimal accuracy in cross-modal scenarios[17],[18].

To address these limitations, we propose SeerRAG, a dynamic-routing RAG framework that follows a Perceive – Shunt – Coordinate paradigm (illustrated in Fig. 1). Our main contributions are threefold:

- a novel two-stage dynamic routing mechanism that unifies uncertainty estimation and modality discrimination to effectively resolve the accuracy – efficiency trade-off inherent in static RAG systems;
- an efficient modular architecture in which the Uncertainty Navigation (UN) module suppresses redundant retrieval through confidence-aware quantification, the Modality Discrimination (MD) engine improves routing precision across modalities, and the Cross-Modal Synergy (CMS) module performs structured alignment of heterogeneous evidence;
- comprehensive validation on the LongDocURL benchmark, demonstrating SeerRAG's generalizability across representative LVLM backbones, including Qwen2-VL-7B and LLaVA-OneVision-Chat-7B, consistent accuracy improvements over state-of-the-art baselines, and the ability to suppress up to 97% of redundant retrieval triggers for factual queries under a hybrid evaluation setting involving both complex and factual queries, thereby confirming the precision of the proposed uncertainty-based gating mechanism.

II. Related Work

A. From LLM Hallucinations to Retrieval-Augmented Generation (RAG)

The reasoning of Large Language Models (LLMs) relies heavily on parametric memory formed during pre-training, making them susceptible to factual hallucinations when processing long-tail knowledge or real-time information. To mitigate this bottleneck, Retrieval-Augmented Generation (RAG) introduces non-parametric knowledge bases[4],[5], establishing a foundation for the integration of vector retrieval and generation[19]. Specifically, FiD[20] enhances multi-document integration through parallel evidence fusion, while Atlas[5] and RETRO[21] demonstrate that retrieval mechanisms enable high knowledge currency with efficient parameter scales. However, these static strategies fail to perceive epistemic boundaries, leading to significant computational redundancy when processing queries solvable by internal model knowledge.

Recent research has pivoted toward enhancing knowledge reliability. FactMM-RAG[22] suppresses hallucinations via multimodal consistency checks; RAG-Check[23] employs a bidirectional consistency verification mechanism to rectify logical contradictions; and RA-CM3[24] incorporates causal reasoning to bolster accuracy in complex tasks. Although these methods optimize generation quality, they remain constrained by static retrieval pipelines and fail to achieve a Pareto-optimal balance between accuracy and efficiency—a gap specifically addressed by the Uncertainty Navigator (UN) module in SeerRAG.

B. Multimodal RAG and Extensions of Visual Retrieval

With the emergence of large vision – language models (LVLMs), such as LLaVA[13],Qwen2-VL[10] ,and Gemini[12] models have gained the ability to directly process multimodal inputs. However, when confronted with evidence containing complex layouts, charts, or extremely long documents, these models remain constrained by limited context window sizes and insufficient multimodal feature alignment capabilities.

To overcome these limitations, multimodal RAG has become an active area of research. Early approaches predominantly relied on optical character recognition (OCR) to convert visual information into textual representations; however, this process inevitably discards critical spatial layouts and visual structural cues. VisRAG[15] addresses this issue by performing feature retrieval directly over page images, demonstrating clear advantages in document localization and visual understanding tasks. Beyond image-based retrieval, graph-structured RAG methods aim to improve retrieval quality through explicit modeling of entity relationships. Among them, HiRAG[25] employs a hierarchical knowledge graph to jointly model semantic and structural information, significantly outperforming traditional GraphRAG approaches [26] in complex domains.

Recent studies indicate a shift in multimodal RAG research from single-modality optimization toward joint adaptation across modalities and granularities. UniversalRAG[17] introduces a modality-aware routing mechanism coupled with multi-granularity retrieval strategies, achieving substantial improvements over unimodal baselines across multiple multimodal benchmarks. Similarly, Taichu-mRAG[27] constructs a unified visual – textual index through fine-grained layout analysis, leading to notable performance gains in document-intensive question answering tasks. Nevertheless, most existing approaches rely on fixed modality routing or static retrieval pipelines, making it difficult to simultaneously balance efficiency and accuracy under heterogeneous query scenarios.

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C. Adaptive Retrieval and Dynamic Routing Mechanisms

To balance generation quality and computational efficiency, adaptive retrieval has emerged as an important research direction in Retrieval-Augmented Generation (RAG). The core idea is to quantify model confidence through uncertainty estimation and dynamically decide whether external retrieval should be triggered. Representative methods such as Self-RAG[14] employ self-reflection mechanisms and output entropy metrics to suppress redundant retrieval, while conditional computation and dynamic routing strategies [28]provide a theoretical foundation for allocating differentiated computation paths to inputs of varying difficulty.

Recent studies have further expanded the modeling scope of adaptive retrieval. SEAKR[29] extracts uncertainty signals from internal hidden states to improve retrieval decisions under ambiguous queries;DeepRAG[30] formulates retrieval decisions as a Markov Decision Process, dynamically optimizing retrieval policies under a benefit–cost trade-off; and OmniSearch[31] supports multi-hop and multimodal queries through question decomposition and planning mechanisms. In parallel, MiniRAG[32] explores lightweight RAG designs for resource-constrained scenarios, extending the applicability of retrieval-augmented generation.

Compared with these approaches, existing methods predominantly focus on whether to retrieve, while largely overlooking the question of which modality to retrieve. SeerRAG addresses this limitation by jointly modeling Uncertainty Navigation (UN) and Modality Discrimination (MD) to construct a two-level dynamic routing mechanism. This design significantly reduces redundant retrieval while preserving generation reliability, providing a unified solution for efficient retrieval-augmented generation in multimodal long-document scenarios.

III. The SeerRAG Framework

A. System Overview

SeerRAG is an adaptive retrieval-augmented generation framework designed for multimodal long-document understanding. As illustrated in Fig. 1, the framework departs from conventional linear retrieval pipelines and instead adopts a hierarchical decision-making architecture based on a Perceive – Route – Coordinate paradigm. By explicitly decoupling retrieval necessity determination from modality-aware routing, SeerRAG achieves a favorable balance between generation accuracy and computational efficiency through three synergistic modules:

- Uncertainty Navigation (UN):Serving as the first-level logical gate, the UN module estimates the knowledge confidence boundaries of the underlying LVLM by jointly modeling confidence scores and sampling consistency. For queries where the model possesses sufficient internal knowledge (e.g., factual queries), SeerRAG performs short-circuit inference to bypass external retrieval and minimize response latency; otherwise, the external knowledge retrieval pipeline is activated.
- Modality Discrimination (MD) Engine:As the second-level routing core, the MD engine integrates query semantics with document layout features to adaptively route each query to textual, visual, or hybrid retrieval paths. This modality-aware scheduling ensures that visually intensive tasks (e.g., chart and figure reasoning) are supplied with necessary image evidence, while avoiding redundant visual computation for text-dominant queries.
- Cross-Modal Synergy (CMS) Generation:Acting as the execution engine, the CMS module performs structured latent-space alignment of heterogeneous retrieved evidence, including textual segments and high-dimensional visual feature representations. By conducting cross-modal consistency modeling within a unified context, this module effectively mitigates information loss in long-document scenarios and substantially improves the factual fidelity of generated responses.

Through this hierarchical strategy — first determining whether retrieval is necessary and then adaptively selecting the appropriate modality path—SeerRAG enables high-precision multimodal reasoning while significantly optimizing end-to-end response efficiency for large-scale heterogeneous query workloads.

B. Uncertainty Navigator (UN)

1) Uncertainty Estimation via Weighted Token Entropy:

For an input query q, the dispersion of the predictive distribution during the autoregressive generation of the LVLM directly reflects the certainty of its internal knowledge. We define the Weighted Token Entropy H_w as the primary metric, as formulated in (1). To emphasize the impact of early-stage semantic tokens on retrieval decisions, a time-step decay coefficient η is introduced:

$$H_w = -\frac{1}{T}\sum_{t=1}^{T} \eta^{t-1} \sum_{v \in V} P(y_t = v | y_{<t}, q) \log P(y_t = v | y_{<t}, q), \quad (1)$$

where $\mathcal{V}$ denotes the vocabulary space, and $P(y_t | y_{<t}, q)$ represents the predictive probability distribution at time-step t. A high entropy value typically signifies that the model encounters epistemic ambiguity at specific semantic nodes, necessitating the intervention of external knowledge.

2) Consistency Verification via MC-Dropout: To further probe the model's epistemic boundaries and suppress "false confidence" caused by training biases, the system employs MC-Dropout technology. By executing K stochastic forward passes with active dropout during inference, we obtain a set of candidate generation sequences. We define the Sampling Consistency Score S_{cons} as the average cosine similarity of these sequences within the semantic embedding space. A low S_{cons} indicates that the model's response to the query is highly unstable, thereby compelling the system to trigger the retrieval logic.

3) Dynamic Decision Criterion: The system generates a final decision score by integrating H_w and S_{cons},producing a composite decision score that is compared against an adaptive threshold τ. The triggering logic $\mathbb{I}_{ret}$ is defined as follows:

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$$\mathbb{I}_{\text{ret}} = \begin{cases} 1, & \text{if Score}(\mathrm{H_w}, \mathrm{S_{cons}}) > \tau \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

when $\mathbb{I}_{ret} = 0$ according to (2), the system determines that the model possesses sufficient parametric knowledge to execute short-circuit inference. This mechanism significantly compresses computational redundancy while maintaining high response precision.

C. Modality Discerner (MD)

Once the Uncertainty Navigation (UN) module determines that external retrieval is required, the Modality Discerner (MD) further analyzes the semantic characteristics of the query to identify the most appropriate retrieval modality.

1) Gated Routing Mechanism: The core of the MD module is a lightweight feature mapping network. The system first extracts dual feature representations of the query q in parallel: a Transformer-based semantic embedding vector f_{txt} and a visual density feature f_{vis} representing the complexity of the document layout. The routing preference factor g as formulated in (3), is calculated via a gating mechanism:

$$\mathrm{g} = \sigma(\mathrm{W_g} \cdot [\mathrm{f_{txt}}; \mathrm{f_{vis}}] + \mathrm{b_g}), \quad (3)$$

where σ denotes the Sigmoid activation function and [·;·]represents the vector concatenation operation. This preference factor determines the dependency weights of the subsequent information flow on textual and visual indices.

2) Path Dispatching and Optimization Objective: According to the distribution of (g), the MD module dispatches each query into one of three discrete routing paths:

- Text Path: Designed for non-visual-dependent tasks such as factual question answering, directing the query to an efficient textual vector index.
- Vision Path: Targeted at queries involving complex chart, figure, or layout reasoning, activating pixel-level visual retrieval.
- Hybrid Path: Intended for cross-modal association tasks in long documents, where textual and visual retrieval are executed in parallel.

To improve modality discrimination accuracy on multimodal long-document benchmarks (e.g., LongDocURL), the MD module is trained using Focal Loss, which encourages the model to focus on hard-to-distinguish modality boundary samples—such as complex statistical charts accompanied by dense textual descriptions. This training strategy enables the MD module to maintain a high modality matching rate under challenging real-world scenarios.

3) Heterogeneous Indexing and Interaction Mechanism: The system maintains synchronized metadata, including page identifiers, image paths, and spatial coordinate offsets. When routing instructions are executed, the metadata management layer ensures that retrieved textual segments and visual regions can be precisely traced to their original sources, providing an interpretable and well-aligned heterogeneous evidence stream for the downstream Cross-Modal Synergy (CMS) module.

D. Cross-Modal Synergist (CMS)

The CMS module serves as the final generation core of SeerRAG, responsible for deeply fusing the heterogeneous evidence retrieved by the MD module and guiding the underlying large vision – language model to perform high-fidelity reasoning. The module follows a two-stage design: (i) structural alignment of heterogeneous evidence and (ii) context-aware synergistic generation.

1) Structural Alignment of Heterogeneous Evidence: The retrieved evidence consists of unstructured textual snippets $\mathcal{E}_{txt}$ and high-dimensional visual pixel regions $\mathcal{E}_{vis}$ .To prevent semantic misalignment,CMS introduces a structured alignment strategy:

- Spatial Feature Anchoring:Beyond raw pixel embeddings, CMS injects spatial coordinate priors derived from layout analysis into the visual representations. By encoding these positional biases, the model can distinguish between functionally identical elements (e.g., two different charts) based on their structural context within the document.
- Semantically Interleaved Encapsulation:CMS utilizes Anchor Tokens to establish deterministic mappings between textual descriptions and corresponding visual segments. This structured encapsulation enforces an implicit feature alignment in the latent space, ensuring that the multimodal context is interleaved in a manner that preserves the original document's logical flow.

*2) Context-Aware Synergistic Generation:*The final generated response y^* follows the principle of maximizing the conditional probability given the query and multi-source evidence, as formalized in (4):

$$y^* = arg\max_{y} \prod_{t=1}^{L} P\left(y_t | y_{<t}, q, \mathcal{A}(\mathcal{E}_{txt}, \mathcal{E}_{vis})\right), \quad (4)$$

where $\mathcal{A}$ denotes the Cross-modal Attention Alignment function. During the decoding phase, this function dynamically computes the relevance weights between the query and multi-source evidence.To further enhance factual grounding, the prompt architecture within CMS embeds Consistency-Checking Constraints. These constraints force the model to cross-reference multiple evidence fragments before generating each token, effectively mitigating the "Lost in the Middle" phenomenon. By prioritizing evidence-grounded reasoning over purely generative completion, CMS significantly reduces factual hallucinations and ensures that the output maintains rigorous cross-modal consistency.

IV. EXPERIMENTS

A. Experimental Setup and Baselines

1) Datasets and Evaluation Tasks: In this study, we select LongDocURL[8] as the primary evaluation benchmark to assess the comprehensive performance of models in multimodal long-document environments. Compared to text-only long-context benchmarks such as LongBench[33], LongDocURL integrates the visual perception capabilities of DocVQA[34] and the logical reasoning requirements of ChartQA[35]. The benchmark

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covers 20 subtasks across over 33,000 document pages, categorized into three core evaluation dimensions: Understanding, Reasoning, and Locating. These tasks examine the model's knowledge acquisition capabilities from multiple perspectives, including semantic, temporal, spatial, and cross-modal dimensions.

2) Implementation Details:

- Encoding and Indexing: As described in Section III-C3, we employ BAAI/bge-large-en[36] as the textual encoder to generate 1024-dimensional dense embeddings. On the visual side, VisRAG is used to extract 768-dimensional visual features from document images.
- Hyperparameter Configuration:In the Uncertainty Navigator (UN) module, the decay factor η is set to 0.9, while the retrieval trigger threshold τ is determined via cross-validation on the validation set. All experiments are conducted on an NVIDIA A800 GPU cluster.

*3) Baseline Methods:*To objectively measure the performance gains of SeerRAG, we compare it against the state-of-the-art results officially reported in the LongDocURL [8] benchmark:

- OCR-based Pipelines: Traditional cascaded workflows that first extract textual content from document pages using OCR tools (e.g., PyMuPDF4), followed by reasoning with text-only large language models. The reported results in this category include representative implementations instantiated with models such as LLaVA-Next-Interleave-7B[37], GPT-4o[38], and o1-preview[39].
- LVLMs (Non-retrieval): Represented by Gemini-1.5-Pro[40] and Qwen-VL-Max[41]. These methods do not rely on retrieval; instead, they perform zero-shot reasoning directly within an ultra-long context window containing document images.
- SeerRAG (Ours): We deploy the proposed SeerRAG framework on two representative LVLM backbones: Qwen2-VL-7B and LLaVA-OneVision-Chat-7B .
- By conducting a rigorous comparison with the official LongDocURL benchmark data, we aim to verify the architectural advantages of SeerRAG in achieving the dual goals of accuracy and efficiency, specifically highlighting the contributions of the UN and MD modules in multimodal routing decisions.

B. Main Results

Table I summarizes the overall performance of SeerRAG on two representative LVLM backbones. The results show that SeerRAG consistently improves multimodal reasoning performance across models with different capability levels, indicating that the proposed framework is not tied to a specific backbone architecture.

For Qwen2-VL-7B, the overall accuracy increases markedly from 30.6% to 67.3%, while for LLaVA-OneVision-Chat-7B, the accuracy improves from 25.0% to 55.1%. These improvements are consistently observed across all major evaluation categories. In particular, substantial gains are achieved in tasks that require long-document comprehension and cross-modal evidence integration, such as Multi-page Understanding and Cross-element Locating. For example, Qwen2-VL-7B achieves an increase from 40.2% to 74.5% on multi-page understanding, demonstrating the effectiveness of SeerRAG in handling complex document structures and distributed evidence.

Table II further analyzes the relationship between Retrieval Accuracy and Final Answer Accuracy. Although SeerRAG attains slightly lower retrieval accuracy than pure vision-based RAG methods in some settings, it consistently yields final answer accuracy. This indicates that maximizing retrieval coverage does not necessarily lead to better reasoning performance. By leveraging Uncertainty Navigation (UN) and Modality Discrimination (MD), SeerRAG selectively retrieves relevant evidence while suppressing redundant or noisy information, resulting in more reliable generation outcomes.

Overall, the experimental results demonstrate that SeerRAG effectively balances reasoning accuracy and retrieval efficiency in multimodal long-document scenarios, highlighting the importance of adaptive retrieval triggering and modality-aware routing in practical RAG systems.

TABLE I. OVERALL PERFORMANCE COMPARISON ON LONGDOCURL BENCHMARK (%)

Model	Understanding		Reasoning		Locating	Total
	single-page	*multi-page*	*single-page*	*multi-page*	*cross-element*	*all*
	612	*631*	*158*	*229*	*695*	*2325*
OCR(PyMuPDF4)+LLMs						
LLaVA-Next-Interleave-7B	19.5	33.1	8.7	23.0	5.8	18.7
Qwen-VL-Max	32.0	43.3	26.8	32.1	27.3	33.3
GPT-4o	30.4	40.2	25.9	30.0	37.2	34.7
O1-preview	31.1	40.1	27.1	34.1	38.6	35.8
Qwen2-VL-7B	24.5	38.1	12.5	29.4	14.9	25.0
LVLMs						
GPT-4o	68.6	68.6	55.7	**62.8**	59.6	64.5
Gemini-1.5-Pro	52.8	58.4	35.9	48.6	46.4	50.9
Qwen-VL-Max	57.7	60.0	44.0	43.8	36	49.5
Qwen2-VL-7B	33.4	40.2	19.3	28.6	22.6	30.6
LLaVA-OneVision-Chat-7B	24.8	36.0	8.0	26.6	18.7	25.0
SeerRAG +LVLMs (Ours)						
SeerRAG+Qwen2-VL-7B	**71.9** ↑	**74.5** ↑	**67.7** ↑	58.1 ↑	**69.2** ↑	**67.3** ↑
SeerRAG+LLaVA-OneVision-Chat-7B	60.0 ↑	65.8 ↑	27.2 ↑	42.4 ↑	51.8 ↑	55.1 ↑

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TABLE II. COMPARISON OF RETRIEVAL ACCURACY AND FINAL SCORE (%)

Model	Understanding		Reasoning		Locating	Total
	single-page 612	*multi-page* 631	*single-page* 158	*multi-page* 229	*cross-element* 695	*all* 2325
Score/Retrieval_Accuracy						
Original RAG(Text)+Qwen2-VL-7B	54.9/85.0	63.9/93.5	44.9/84.2	43.2/87.3	50.7/86.6	54.2/82.8
Original RAG(Text)+LLaVA-OneVision-Chat-7B	55.6/85.0	56.7/93.5	25.3/84.2	37.5/87.3	46.6/86.6	49.4/82.8
Original RAG(VisRAG)+Qwen2-VL-7B	72.9/97.1	68.8/98.1	60.1/95.6	55.5/98.3	62.9/95.4	66.2/96.9
Original RAG(VisRAG)+LLaVA-OneVision-Chat-7B	37.6/97.1	48.9/98.1	19.6/95.6	34.9/98.3	41.4/95.4	40.3/96.9
SeerRAG+Qwen2-VL-7B	71.9/93.6	74.5/97.6	67.7/94.3	58.1/94.8	69.2/95.4	67.3/95.4
SeerRAG+LLaVA-OneVision-Chat-7B	60.0/93.6	65.8/97.6	27.2/94.3	42.4/94.8	51.8/95.4	55.1/95.4

C. Ablation Study and Empirical Analysis

To quantitatively evaluate the individual contributions of the Uncertainty Navigation (UN) module and the Modality Discerner (MD) to the overall system performance, we conduct a series of controlled experiments. We construct a hybrid evaluation set consisting of 500 complex cross-modal queries from LongDocURL (requiring retrieval) and 500 factual queries from ARC-Easy[42] (amenable to direct generation) to examine the robustness of the framework across varying query difficulties.

1) Experimental Configurations: We evaluate four distinct configurations to isolate the impact of each component:

- Original RAG (Text/VisRAG): The baseline configuration where a single-modality retrieval process is forcibly triggered for every query regardless of necessity.
- UN Only: This setup only activates the retrieval necessity determination; if retrieval is deemed necessary, a fixed single-modality engine is invoked.
- MD Only: This configuration employs the modality recognition logic for all queries but bypasses the uncertainty-based filtering, executing retrieval for the entire dataset.
- SeerRAG (Full): The complete two-stage routing mechanism integrating both the UN and MD modules.

2) Results Performance Analysis: The experimental results are summarized in Table III and visualized in Fig. 2 and Fig. 3. Based on the comparative analysis, we draw the following conclusions:

Efficiency Gains via Uncertainty Navigation (UN): Comparing the Original RAG with the UN-only configuration reveals that the UN module effectively identifies simple queries that do not require external knowledge. Specifically, for UN-only (Text), the average generation latency dropped from 7.4s to 3.2s, resulting in a 46% reduction in end-to-end latency. This validates that "on-demand retrieval" based on query confidence is a critical factor in increasing the throughput of large-scale reasoning systems.

Accuracy Improvements via Modality Discrimination (MD): For complex tasks involving intricate layouts or chart reasoning (e.g., LongDocURL), the MD-only configuration demonstrated significant superiority, achieving a total accuracy of 78.0%, compared to 69.5% for the text-only RAG. The MD module accurately identifies vision-intensive queries and switches to the visual retrieval path. However, without the filtering of the UN module, the average total latency of MD-only (10.6s) remains considerably higher than the full system, underscoring the necessity of hierarchical routing.

Pareto Optimality of SeerRAG: As shown in the Accuracy-Efficiency Pareto Frontier (Fig. 2), the full SeerRAG (UN+MD) configuration successfully occupies the optimal upper-left region of the performance space. While SeerRAG introduces a minor routing overhead of approximately 0.88s (as detailed in Fig. 3), it achieves the highest total accuracy of 79.0% and reduces the total latency to 7.6s compared to the 17.3s of the Original (VisRAG) baseline, yielding a 2.3 × end-to-end speedup.

TABLE III. ABLATION STUDY RESULTS

Configuration	Accuracy (%)	Retrieval (ms)	Routing (s)	Generation (s)	Total Latency (s)
Original RAG (Text)	69.5	35	7.4	0	7.4
Original RAG (VisRAG）	77.3	23	17.3	0	17.3
UN only (Text)	70.8	19	3.2	0.77	4.0
UN only (VisRAG)	78.4	12	8.9	0.77	9.682
MD only	78.0	34	10.4	0.21	10.6
SeerRAG (Ours)	79.0	17	6.7	0.88	7.6

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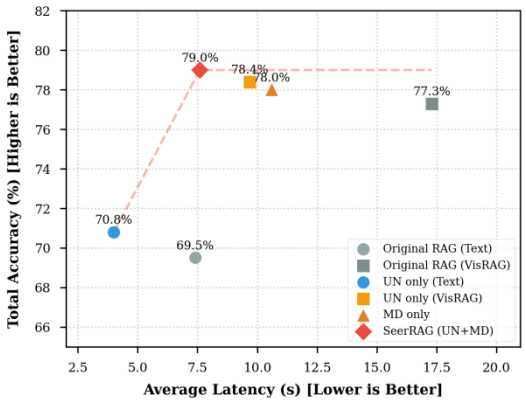


Fig. 2. Accuracy-Efficiency Pareto Frontier on the Mixed Dataset. Points closer to the upper-left corner represent better overall performance, with the dashed red line indicating the Pareto optimal boundary.

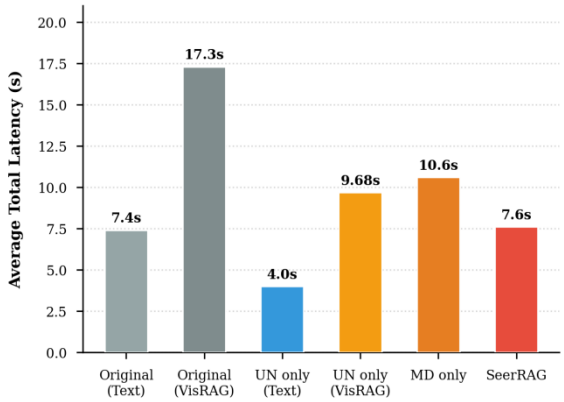


Fig. 3. Decomposition of Average Total Latency across different configurations.

V. Conclusion and Future Work

This paper addresses two critical limitations of existing multimodal retrieval-augmented generation (RAG) systems, namely retrieval redundancy and rigid modality adaptation. We propose SeerRAG, a dynamic retrieval-augmented generation framework guided by uncertainty estimation and modality-aware routing. By jointly integrating the Uncertainty Navigation (UN), Modality Discrimination (MD), and Contextual Management Strategy (CMS) modules, SeerRAG enables fine-grained multimodal routing and efficient evidence utilization for complex long-document queries.

Extensive experiments on the LongDocURL benchmark demonstrate that SeerRAG consistently improves long-document reasoning performance across different LVLM backbones. When applied to Qwen2-VL-7B and LLaVA-OneVision-Chat-7B, SeerRAG achieves substantial accuracy gains compared to their zero-shot counterparts, with improvements exceeding 100% in overall long-document understanding accuracy. In addition, SeerRAG effectively suppresses redundant retrieval, leading to better evidence relevance and stronger end-to-end reasoning reliability. These results validate the effectiveness of dynamic routing mechanisms in multimodal long-document scenarios and highlight the robustness and generality of the proposed modular design.Despite its strong performance, SeerRAG has several limitations. First, modality discrimination relies on pretrained feature extractors, which may exhibit reduced robustness under low-quality document scans or highly entangled cross-modal semantics. Second, the scalability of cross-modal indexing and real-time retrieval remains challenging when extending to ultra-large (e.g., TB-scale) multimodal knowledge bases.

Future work will focus on three directions: (1) improving routing robustness through adaptive uncertainty threshold learning and end-to-end joint optimization of UN, MD, and CMS;(2) extending the framework to additional modalities such as audio, video, and 3D data, while optimizing hierarchical indexing for long and streaming documents; and (3) validating SeerRAG in high-impact applicationdomains, including medical decision support, financial compliance analysis,academic literature understanding, and intelligent manufacturing systems.

Overall, SeerRAG provides a principled and scalable solution for efficient and reliable multimodal RAG, offering a promising foundation for next-generation knowledge-intensive AI systems.

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References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, "Attention is all you need," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2017, pp. 5998–6008.

[2] J. Achiam, S. Adler, S. Agarwal, J. Berner, C. Chan, F. Gao, P. Gehrmann, C. Hallacy, J. He, T. Henighan, B. Hunter, B. Leskovec, P. Mishkin, K. Montague, H. Edwards, J. Welinder, P. Christiano, I. Sutskever, "GPT-4 technical report," arXiv preprint arXiv:2303.08774, 2023.

[3] Z. Ji, N. Lee, R. Frieske, Y. Yu, S. Su, D. Xu, Y. Feng, X. Cambria, N. S. Balasubramanian, P. Fung, "Survey of hallucination in natural language generation," ACM Comput. Surv. (CSUR), vol. 55, no. 12, pp. 1-38, 2023.

[4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020, pp. 9459-9474.

[5] Y. Gao, Y. Xiong, G. Gao, X. Li, Y. Lyu, P. Shum, Y. Chi, "Retrieval-augmented generation for large language models: A survey," arXiv preprint arXiv:2312.10997, 2023.

[6] G. Izacard, P. Lewis, M. Lomeli, L. von Werra, A. Malaviya, S. Patil, F. Petroni, E. Grave, "Atlas: Few-shot learning with retrieval augmented language models," J. Mach. Learn. Res. (JMLR), vol. 24, no. 1, pp. 1-43, 2023.

[7] W. Chen, H. Hu, C. Saharia, W. W. Cohen, "MuRAG: Multimodal retrieval-augmented generation with pre-trained visual-language models," in Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP), 2022, pp. 1-12.

[8] C. Deng, J. Yuan, P. Bu, Y. Liu, J. Wang, H. Li, "LongDocURL: A comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating," in Proc. 63rd Annual Meeting of the Assoc. Comput. Linguist. (ACL), 2025, pp. 1135-1159.

[9] Y. Liu, J. Wang, Z. Li, "Multimodal long-document retrieval with layout-aware embedding," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2024, pp. 10245-10254.

[10] B. Peng, X. Zhu, H. Cui, Y. Song, J. Geng, X. Liu, Y. Gao, J. Su, Y. Piao, Q. Wang, et al., "Qwen2-VL: To see the world from a better perspective," arXiv preprint arXiv:2410.02719, 2024.

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[11] B. Li, Y. Zhang, D. Guo, R. Zhang, F. Li, H. Zhang, K. Zhang, Y. Li, Z. Liu, C. Li, "LLaVA-OneVision: Easy visual task transfer," arXiv preprint arXiv:2408.03326, 2024.

[12] M. Reid, N. Savinov, D. Teplyashin, A. Ahuja, A. M. Dai, S. Norouzi, J. Lee, "Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context," arXiv preprint arXiv:2403.05530, 2024.

[13] H. Liu, C. Li, Q. Wu, P. Wang, Y. J. Lee, Y. Gao, J. Bu, J. Gao, L. Zhang, "Visual instruction tuning," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2023.

[14] A. Asai, Z. Wu, Y. Wang, A. Sil, H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in Proc. Int. Conf. Learn. Represent. (ICLR), 2024.

[15] W. Yu, Z. Zhu, S. Hou, X. Wang, Y. Cao, J. Huang, L. Zhang, J. Gao, "VisRAG: Vision-based retrieval-augmented generation," arXiv preprint arXiv:2408.01314, 2024.

[16] J. Chen, H. Lin, X. Han, L. Sun, "Benchmarking large language models in retrieval-augmented generation," in Proc. AAAI Conf. Artif. Intell. (AAAI), 2024, pp. 17754–17762.

[17] X. Wang, Y. Li, Z. Chen, J. Wang, H. Li, "UniversalRAG: Retrieval-augmented generation across modalities and granularities," in Proc. Int. Conf. Mach. Learn. (ICML), 2025, pp. 12456–12471.

[18] S. Zhang, H. Chen, X. Wu, "Adaptive modality routing for real-time multimodal RAG," in Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP), 2025, pp. 7890–7894.

[19] K. Guu, K. Lee, Z. Tung, P. Pasupat, M. Chang, "REALM: Retrieval-augmented model pre-training," in Proc. Int. Conf. Mach. Learn. (ICML), 2020, pp. 3929–3938.

[20] G. Izacard, E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," arXiv preprint arXiv:2007.01282, 2020.

[21] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, G. Balwit, A. Bales, Y. Huang, A. Bapna, et al., "Improving language models by retrieving from trillions of tokens," in Proc. Int. Conf. Mach. Learn. (ICML), 2022, pp. 2206–2240.

[22] X. Sun, Y. Li, Z. Wang, Y. Liu, J. Zhou, M. Sun, "FactMM-RAG: Fact-aware multimodal retrieval-augmented generation for reliable reasoning," in Proc. AAAI Conf. Artif. Intell. (AAAI), 2024, pp. 12345–12353.

[23] A. Mortaheb, S. Zhang, H. Hajishirzi, "RAG-Check: Verifying retrieval-augmented generation outputs via evidence consistency," arXiv preprint arXiv:2501.03456, 2025.

[24] K. Yasunaga, J. Chen, P. Liang, "RA-CM3: Retrieval-augmented causal reasoning for multimodal models," in Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP), 2023, pp. 4567–4580.

[25] H. Huang, Y. Zhang, J. Liu, "HiRAG: Hierarchical knowledge-enhanced graph retrieval-augmented generation," arXiv preprint arXiv:2503.10150v1, 2025.

[26] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, J. Larson, "From local to global: A graph RAG approach to query-focused summarization," arXiv preprint arXiv:2404.16130, 2024.

[27] J. Zhang, Y. Liu, H. Wang, X. Li, Z. Wang, "Taichu-mRAG: Fine-grained multimodal retrieval-augmented generation for rich document understanding," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2025, pp. 15678–15688.

[28] W. Fedus, B. Zoph, N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," J. Mach. Learn. Res. (JMLR), vol. 23, no. 120, pp. 1–39, 2022.

[29] J. Kim, S. Lee, H. Park, J. Kim, "SEAKR: Self-aware knowledge retrieval for adaptive retrieval-augmented generation," in Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP), 2025, pp. 8901–8915.

[30] Y. Chen, Z. Huang, J. Zhao, H. Zhang, "DeepRAG: Markov decision process for optimized retrieval in augmented generation," in Proc. AAAI Conf. Artif. Intell. (AAAI), 2025, pp. 13890–13898.

[31] A. Liu, M. Zhang, Y. Chen, X. Wang, "OmniSearch: Adaptive planning agent for dynamic multimodal retrieval-augmented generation," in Proc. ACM Multimedia, 2025, pp. 4567–4578.

[32] Z. Hong, X. Li, M. Chen, "MiniRAG: Towards extremely simple retrieval-augmented generation," arXiv preprint arXiv:2501.06713v3, 2025.

[33] Y. Bai, X. Lv, J. Zhang, Y. Su, J. Liu, T. Liu, X. Wang, P. Li, "LongBench: A comprehensive, multilingual benchmark for long context understanding," in Proc. 61st Annual Meeting of the Assoc. Comput. Linguist. (ACL), 2023.

[34] M. Mathew, D. Karatzas, C. V. Jawahar, "DocVQA: A dataset for VQA on document images," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2021, pp. 2200–2209.

[35] A. Masry, D. Xuan Long, J. Q. Tan, S. Joty, E. Hoque, "ChartQA: A benchmark for question answering about charts with visual and logical reasoning," in Proc. 60th Annual Meeting of the Assoc. Comput. Linguist. (ACL), 2022, pp. 2263–2279.

[36] S. Xiao, Z. Liu, P. Zhang, Y. Dong, H. Wang, and F. Wei, "BGE: Towards better general embedding with multi-stage contrastive learning and knowledge enhancement," arXiv preprint arXiv:2312.05772, 2023.

[37] F. Li, R. Zhang, H. Zhang, Y. Zhang, B. Li, W. Li, Z. Ma, and C. Li, "LLaVA-NeXT-Interleave: Tackling Multi-image, Video, and 3D in Large Multimodal Models," arXiv preprint arXiv:2407.07895, 2024.

[38] OpenAI, "Hello GPT-4o," OpenAI Official Blog, May 13, 2024. [Online]. Available: https://openai.com/research/gpt-4o [Accessed: Dec. 29, 2025].

[39] OpenAI, "Introducing OpenAI o1-preview," OpenAI Blog, September 12, 2024. [Online]. Available: https://openai.com/index/introducing-openai-o1-preview/ [Accessed: Dec. 29, 2025].

[40] M. Reid, N. Savinov, D. Teplyashin, D. J. Cole, M. W. Hoffman, W. M. Lan, et al., "Gemini 1.5: Unlocking multimodal understanding across millions of tokens," arXiv preprint arXiv:2403.05531, 2024.

[41] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, et al., "Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond," arXiv preprint arXiv:2308.12966, 2023.

[42] J. Clark, N. Luong, Q. Le, C. Finn, and S. Riedel, "Think you have solved question answering? Try ARC, the AI2 Reasoning Challenge," in Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP), 2018, pp. 1822–1832.

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